be less than the escape velocity from the planet if the gases should not evaporate into space. The escape velocity depends of temperature and mass of the planet (cf. §12.1.2); for example, the moon has too small a mass to keep any atmosphere. The escape velocity of H_2 and He from the Earth is too large for these elements to be retained. The original atmosphere of the Earth is therefore believed to have been mainly H_2O, CO_2, N_2, CH_4 and NH_3. This is referred to as a *reducing atmosphere*, which probably is typical for all newborn Earth-sized planets. The UV-radiation from the sun decomposes these molecules (though the fragments may recombine), releasing H_2, which escapes. The O_2 released is very reactive and combines with many elements to form oxides; most of the red surface of Mars is FeO. The common assumption is that very little oxygen was originally present as free O_2. However, rather rapidly, processes led to a shift of the atmosphere from reducing towards *oxidizing conditions*. Life appears to have begun before this transition as evidenced by fossile bacteria-like organisms in very old rocks formed in a reducing atmosphere.

12.8. EXERCISES

12.1. (a) What is the most probable kinetic energy of a hydrogen atom at the interior of the sun ($T = 1.5 \times 10^7$ K)? (b) What fraction of the particles would have energies in excess of 100 keV?

12.2. What amount (kg) of hydrogen is consumed per second by the sun in the fusion reaction 12.6 in Table 12.2? The solar constant (energy flux at earth's orbit) is 1.353 kW m^{-2} and the earth's average distance to the center of the sun is 149 599 000 km.

12.3. Assuming the sun generates its energy to 2/3 by the proton-chain and the rest by the CNO cycle, what would the expected solar neutrino flux be at the center of the earth?

12.4. In a water power station with a fall height of 20 m and a water flow of 500 $m^3\ s^{-1}$, the electric power output at 100% efficiency can be calculated. If the heavy water in the fall could be extracted and converted in a fusion reactor according to (12.14) with a 25% efficiency, which of the two power sources would yield more energy?

12.5. The supernova SN1987A was at a distance of about 51.47 kPc from us and had about 10^{46} J total energy release as estimated from the neutrino flux hitting Earth. How much mass in kg was then converted to energy in the explosion?

12.9. LITERATURE

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